

BOLNISI, GeorgiaWMO: **376210**Lat: **41.4509N**Lon: **44.5585E**Elev: **1762**StdP: **13.78**Time Zone: **4.00 (E04)**Period: **05-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	20.2	23.7	8.3	9.0	27.9	12.1	10.8	28.7	13.2	47.3	9.0	38.1	0.6	270	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	18.7	93.7	74.2	91.4	72.7	89.2	71.6	77.4	89.5	75.3	88.5	73.4	86.5	2.3	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
74.0	135.2	86.2	71.1	122.5	84.1	69.0	113.9	82.3	42.4	89.3	40.2	88.3	38.3	86.7	85.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.1	6.1	4.6	DB	17.8	97.3	5.5	2.3	13.8	99.0	10.6	100.4	7.5	101.7	3.4	103.4	(4)
(5)			WB	15.5	78.6	4.9	3.3	12.0	81.0	9.2	82.9	6.4	84.7	2.9	87.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	56.8	36.3	38.6	46.5	54.4	63.9	72.9	77.6	77.7	68.8	58.0	46.0	39.2	(6)	
	DBStd	15.91	6.33	6.97	6.00	6.72	5.44	5.03	4.84	5.29	6.04	5.97	6.38	5.64	(7)	
	HDD50	1417	426	321	142	32	1	0	0	0	0	11	147	337	(8)	
	HDD65	4240	890	739	574	321	85	3	1	1	30	226	570	800	(9)	
	CDD50	3886	1	3	33	164	431	687	857	860	563	258	27	2	(10)	
	CDD65	1234	0	0	0	4	50	240	393	396	143	8	0	0	(11)	
	CDH74	11855	0	0	1	35	329	2050	4051	4290	1061	38	0	0	(12)	
	CDH80	4833	0	0	0	2	41	695	1795	1999	299	2	0	0	(13)	
	WSAvg	1.6	1.3	1.6	2.0	1.6	1.7	1.9	1.9	2.0	1.6	1.4	1.3	1.3	(14)	
	PrecAvg	22.7	0.9	1.1	1.7	3.0	3.3	3.2	1.7	1.5	1.8	2.2	1.5	0.9	(15)	
(16) Precipitation	PrecMax	28.9	2.1	2.5	3.9	5.8	5.6	6.1	3.5	4.3	4.9	5.4	3.7	3.1	(16)	
	PrecMin	17.8	0.1	0.2	0.1	0.9	1.1	1.1	0.6	0.2	0.4	0.6	0.1	0.0	(17)	
	PrecStd	3.2	0.6	0.6	1.0	1.3	1.1	1.3	0.8	0.9	1.1	1.1	1.0	0.7	(18)	
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	56.1	62.7	68.3	78.0	84.4	92.5	97.1	97.1	90.1	78.0	65.7	59.1	(19)	
		MCWB	44.3	47.8	56.8	64.8	67.1	71.3	75.8	76.5	72.0	63.7	53.4	46.7	(20)	
	2%	DB	52.1	57.0	64.0	73.8	80.1	89.4	93.6	94.2	86.7	73.9	62.0	54.5	(21)	
		MCWB	41.9	45.6	52.7	60.5	65.0	70.8	74.0	75.0	70.6	60.9	51.0	44.3	(22)	
	5%	DB	48.7	52.8	60.5	70.4	77.7	86.7	91.2	92.1	83.5	70.7	58.4	51.1	(23)	
		MCWB	39.7	42.9	50.1	58.7	64.4	70.2	73.5	73.0	68.2	58.9	49.1	42.6	(24)	
	10%	DB	45.3	49.1	57.3	66.8	74.9	84.2	88.9	89.9	80.3	67.9	55.4	48.0	(25)	
		MCWB	37.8	40.8	48.0	56.3	62.8	69.0	72.0	71.5	66.4	57.6	47.7	41.0	(26)	
	0.4%	WB	45.6	48.6	60.0	66.8	71.5	77.2	82.0	79.8	74.7	65.3	55.3	48.9	(27)	
		MCDB	55.2	60.4	67.2	76.0	81.4	85.2	89.0	92.8	86.1	74.0	61.6	55.7	(28)	
(30) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	42.6	46.1	53.9	62.5	67.5	74.0	77.9	77.4	71.7	62.5	52.9	45.9	(29)	
		MCDB	49.4	55.6	60.9	70.9	76.5	84.9	88.7	90.7	84.3	70.9	59.0	52.2	(30)	
	5%	WB	40.7	43.5	50.9	59.4	65.3	71.7	75.4	75.4	69.5	60.5	50.9	43.9	(31)	
		MCDB	46.9	51.8	58.8	69.1	75.1	84.2	88.6	89.5	81.1	68.2	55.9	49.4	(32)	
	10%	WB	38.7	41.3	48.6	56.9	63.4	69.8	73.6	73.3	67.5	58.6	49.0	41.9	(33)	
		MCDB	44.2	47.8	56.1	65.6	73.3	82.2	86.8	87.0	78.2	65.9	54.1	47.2	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	12.9	13.5	15.7	16.1	16.7	18.1	18.4	18.7	16.8	14.1	13.3	13.1	(35)	
		MCDBR	17.6	20.5	21.0	21.3	21.0	20.9	21.1	21.8	20.7	18.9	18.1	17.2	(36)	
		MCWBR	11.6	13.0	13.5	12.6	11.5	10.2	10.4	11.4	10.7	10.2	11.0	11.1	(37)	
	5% WB	MCDBR	15.5	18.7	18.8	20.5	19.1	20.0	19.8	20.8	19.3	16.7	15.0	14.5	(38)	
		MCWBR	10.6	12.1	12.6	12.7	11.4	11.5	11.0	11.8	10.9	9.4	9.9	9.9	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.327	0.366	0.417	0.485	0.453	0.441	0.484	0.479	0.423	0.402	0.349	0.324	(40)	
	taud		2.320	2.197	2.094	1.955	2.120	2.217	2.109	2.117	2.258	2.244	2.351	2.349	(41)	
	Ebn at Noon		257	261	260	250	263	266	252	250	257	246	246	247	(42)	
	Edh at Noon		28	36	46	56	49	44	49	47	38	35	27	25	(43)	
(44) All-Sky Solar Radiation	RadAvg		607	908	1111	1376	1663	1959	1913	1755	1374	913	643	512	(44)	
	RadStd		74	137	120	178	115	143	159	187	113	100	78	67	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (20 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page